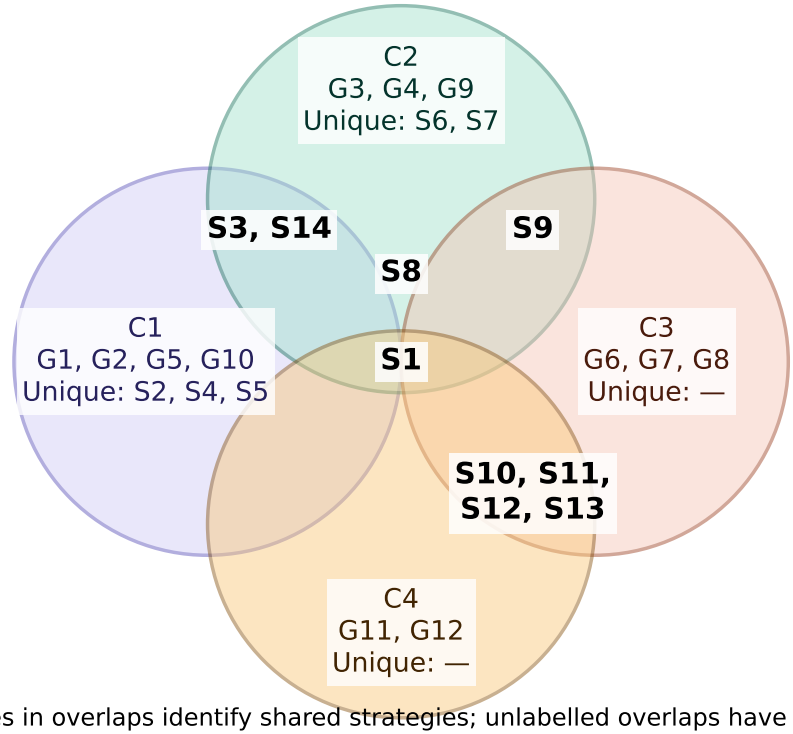

Codes in overlaps identify shared strategies; unlabelled overlaps have none.

Research gaps

C1 Data Infrastructure

G1 Absence of public benchmarks

G2 Generalizability & domain shift

G5 Annotation scarcity

G10 Privacy-preserving & multi-institutional learning

C2 Modeling & Learning

G3 Temporal modeling & video-level analysis

G4 2D constraints & 3D reconstruction

G9 Class imbalance & rare pathology coverage

C3 Clinical Deployment

G6 Clinical validation

G7 Explainability & model transparency

G8 Multi-modal data integration

C4 Outcome & Reporting

G11 Clinical outcome prediction

G12 LLMs & structured reporting

Mitigation strategies

S1 Large-scale multi-institutional annotated angiographic datasets

S2 Dedicated challenge benchmarks with standardized evaluation metrics

S3 Self-supervised / foundation model pretraining on unlabeled fluoroscopic video

S4 Federated learning with data-use agreements & audit mechanisms

S5 Domain adaptation for contrast, dose, frame rate, and motion

S6 Video Transformer architectures with long-range attention & SSMs

S7 Physics-informed networks with projection geometry & multi-view fusion

S8 Semi-supervised, active & self- supervised learning for annotation

S9 Uncertainty quantification & OOD failure detection

S10 Prospective multi-center RCTs with clinical outcome endpoints

S11 Outcome-supervised training for clinically meaningful prediction

S12 Longitudinal cohort datasets linking imaging to clinical endpoints

S13 Vision-language models for image & free-text report integration

S14 Generative augmentation & rare-event simulation pipelines